

ENVS193_Final

Caleb Neely

Set Up

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
nests <- read_csv("nest_data_final.csv")
```

Rows: 299 Columns: 10

-- Column specification -----

```
Delimiter: ","
chr (3): alive, bird_species, ant_species
dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Problem 1: Research writing

a) Transparent statistical methods

Part 1: The co-worker likely used a logistic regression because the response variable is the probability of American Avocet microhabitat use (used vs. not used) and the predictor variable is distance to the water edge, which is continuous.

Part 2: The co-worker likely used a chi-square test of independence because both bird species and habitat type are categorical variables, and the goal is to determine whether the distribution of bird species differs among habitat types.

b) Suggestions For Rewriting

Part 1: American Avocets were more likely to use some areas than others depending on their distance from the water's edge, suggesting that proximity to water influences microhabitat selection.

Part 2: The three bird species did not use habitats in the same way. Habitat use differed among species, indicating that each species may prefer different habitat types within the San Francisco Bay.

c) Further Analysis

One additional variable that could influence the probability of American Avocet microhabitat use is water depth (cm), measured at each survey location using a meter stick or depth gauge. Water depth may affect habitat use because American Avocets forage by sweeping their bills through shallow water, so they may prefer areas with depths that maximize feeding efficiency.

Problem 2 Data analysis

a) Clean and Wrangle

```
nests_clean <- nests |>
  select(case_control, height_cm, ant_species) |>
  filter(ant_species %in% c("AB", "BBR", "RRB")) |>
  mutate(
    ant_species = factor(
      ant_species,
      levels = c("BBR", "AB", "RRB"),
      labels = c(
        "C. sjostedti",
        "C. mimosae",
        "C. nigriceps"
      )
    ),
    ant_species_full = case_when(
      ant_species == "C. sjostedti" ~ "Crematogaster sjostedti",
      ant_species == "C. mimosae" ~ "Crematogaster mimosae",
      ant_species == "C. nigriceps" ~ "Crematogaster nigriceps"
    )
  )

str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control      : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm        : num [1:270] 392 310 513 154 324 ...
 $ ant_species       : Factor w/ 3 levels "C. sjostedti",...: 3 3 3 3 2 3 1 3 2 1 ...
 $ ant_species_full  : chr [1:270] "Crematogaster nigriceps" "Crematogaster nigriceps" "Crematogaster nigriceps" ...
```

```
nests_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
  case_control height_cm ant_species ant_species_full
      <dbl>      <dbl> <fct>          <chr>
1           0       236 C. nigriceps  Crematogaster nigriceps
2           1       238 C. nigriceps  Crematogaster nigriceps
```

3	0	141	C. sjostedti	Crematogaster sjostedti
4	0	168.	C. nigriceps	Crematogaster nigriceps
5	1	166.	C. sjostedti	Crematogaster sjostedti
6	1	166	C. nigriceps	Crematogaster nigriceps
7	1	392.	C. nigriceps	Crematogaster nigriceps
8	0	270.	C. nigriceps	Crematogaster nigriceps
9	0	353	C. nigriceps	Crematogaster nigriceps
10	0	266	C. nigriceps	Crematogaster nigriceps

b) Response variable

The response variable is nest occurrence, where a value of 1 indicates that a bird nest was present in a tree and a value of 0 indicates that no bird nest was present. Biologically, this variable represents whether a tree was selected and used as a nesting site by birds.

c) Purpose of Study

Tree height:

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase because taller trees can provide better nesting locations and may be more visible or accessible to birds while also reducing the risk of some predators. This information is discussed in the Methods section, “Nest Site Selection” subsection, where tree characteristics were measured, and is supported by the Results section, where tree height was included as a predictor of nest occurrence.

Acacia ant species:

Acacia ant species is a categorical variable with three levels: *C. sjostedti*, *C. mimosae*, and *C. nigriceps*. The probability of nest occurrence may differ among ant species because birds preferentially nest in trees occupied by more aggressive ant species that defend trees against browsing mammals and potential nest predators. This information is described in the Introduction (final paragraph) and supported in the Discussion section, where the authors explain that birds selected trees inhabited by the more aggressive ant species.

d) Table of Models

```
model_table <- tibble(
  model_number = c("Model 1", "Model 2"),
  response_variable = c(
    "Nest occurrence",
    "Nest occurrence"
  ),
```

```

predictors = c(
  "Tree height + acacia ant species",
  "Tree height + acacia ant species + tree height × acacia ant species"
),
interaction_term = c(
  "No",
  "Yes"
)
)

model_table

```

```

# A tibble: 2 x 4
  model_number response_variable predictors interaction_term
  <chr>         <chr>           <chr>           <chr>
1 Model 1      Nest occurrence Tree height + acacia ant spec~ No
2 Model 2      Nest occurrence Tree height + acacia ant spec~ Yes

```